

A PROJECT TITLE

ON

**Credit Card Fraud Detection Using Logistic Regression and Isolation-Forest Anomaly
Detection**

A project report submitted in partial fulfillment of the requirements for the Course
MACHINE LEARNING

BACHELOR OF TECHNOLOGY

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by

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DEPARTMENT OF Computer Science and Engineering

Name of Title: Credit Card Fraud Detection Using Logistic Regression and Isolation-Forest Anomaly

1. Abstract

Credit card fraud has become a major concern in the rapidly expanding digital payment ecosystem, leading to significant financial losses for both financial institutions and customers. Detecting fraudulent transactions is challenging due to the highly imbalanced nature of transaction datasets, where genuine transactions vastly outnumber fraudulent ones. This project presents a hybrid fraud detection approach that combines **Logistic Regression**, a supervised machine learning algorithm, with **Isolation Forest**, an unsupervised anomaly detection technique. Logistic Regression is employed to classify transactions as fraudulent or legitimate based on historical labeled data by learning the relationship between transaction features and fraud labels. In contrast, Isolation Forest identifies unusual transaction patterns by isolating anomalies without requiring labeled data, making it effective for detecting previously unseen fraud types. The combination of these two techniques improves the overall robustness of the detection system by leveraging both predictive classification and anomaly detection capabilities. The proposed model includes data preprocessing, feature scaling, handling class imbalance, model training, and performance evaluation using metrics such as accuracy, precision, recall, F1-score, and ROC-AUC. Experimental results demonstrate that the hybrid approach achieves better fraud detection performance than using either method individually, particularly in identifying rare fraudulent transactions while minimizing false positives. This approach offers a scalable and efficient solution for real-time credit card fraud detection, helping financial organizations enhance transaction security and reduce financial risks.



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2. INTRODUCTION

The rapid growth of digital payment systems and online banking has significantly increased the use of credit cards for financial transactions. While this has improved convenience and accessibility, it has also led to a rise in fraudulent activities. Credit card fraud results in substantial financial losses for banks, merchants, and customers, making accurate and timely fraud detection an essential requirement in the financial industry.

Traditional fraud detection methods rely heavily on manually defined rules and expert knowledge. However, these approaches struggle to detect new and evolving fraud patterns. Machine Learning (ML) provides a more effective solution by learning patterns from historical transaction data and identifying suspicious activities automatically.

This project focuses on developing a **Credit Card Fraud Detection System** using two complementary machine learning techniques: **Logistic Regression** and **Isolation Forest**. Logistic Regression is a supervised classification algorithm that predicts whether a transaction is fraudulent based on labeled training data. Isolation Forest is an unsupervised anomaly detection algorithm that identifies unusual transactions by isolating outliers, making it capable of detecting previously unseen fraud patterns.

The project utilizes a publicly available **Credit Card Fraud Detection dataset from Kaggle**, which contains anonymized transaction features, transaction amount, transaction time, and class labels indicating whether a transaction is legitimate or fraudulent. Since fraudulent transactions represent only a small fraction of the dataset, appropriate preprocessing techniques such as feature scaling and class imbalance handling are applied to improve model performance.

The models are implemented using **Python** and evaluated using performance metrics including Accuracy, Precision, Recall, F1-Score, and ROC-AUC. The proposed system aims to provide an efficient, accurate, and scalable solution for identifying fraudulent credit card transactions while reducing false alarms and improving financial security.



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3. SOFTWARE REQUIREMENTS

The development and evaluation of the match outcome prediction system rely on a Python based machine learning ecosystem. The table below summarizes the software components used across data preprocessing, model building, evaluation, and reporting stages of the project.

Software requirements	Software used
Operating Systems	Windows 10/11
Programming Language	Python
Development Environment	Google Colab or Visual Studio Code
DataSet	Credit Card Fraud Detection Dataset from Kaggle (.csv format) https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud?resource=download


Signature of the Guide


Course Instructor